Department of Mechanical Engineering

Mechanical Engineering Design

ENME 301 Assignment 1: Aluminium Structure

Coversheet for online submission

AUTHORSHIP	
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DECLARATION (MUST BE SIGNED):	
I (we, if group work) have read and fully understood the Department's Statement Regarding Dishonest Practice (on the back) and hereby certify that this item of work submitted for assessment is entirely my/our own work.	
Signed:	Signed:
Date: 25/03/2024	Date: 25/03/2024

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(in connection with work submitted for assessment)

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Introduction

This report covers the design development of a load bearing aluminium structure. The requirements of this structure are that it must support a mass of 20 kg and fail under a mass less than or equal to 39 kg. The structure must be supported by a roller and pin joint and fit within the dimensions specified in Figure 1. To manufacture the support, only aluminium strips, steel pins, and epoxy resin are to be used.

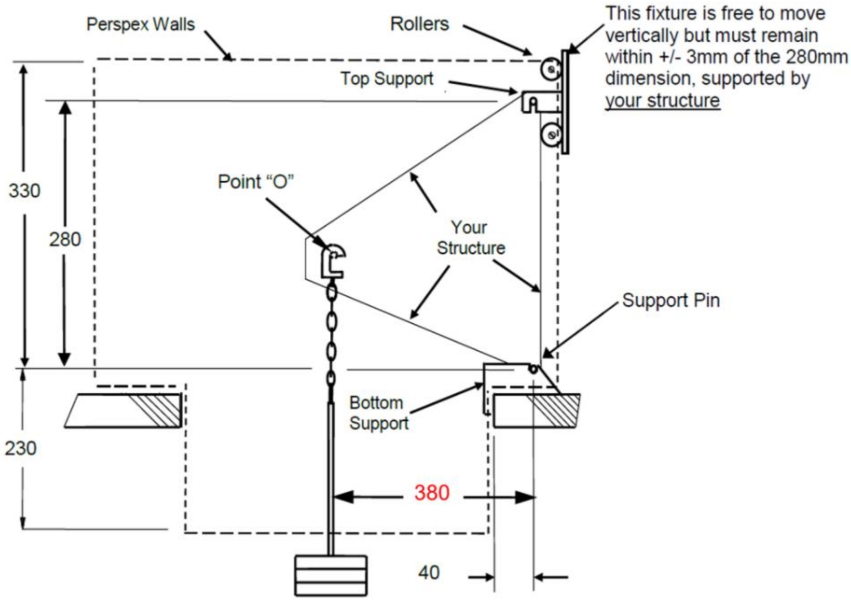


Figure 1: Diagram of the testing apparatus.

Testing and Results

The grade of aluminium used in manufacturing the support is unknown. To determine its material properties, tensile testing was required. Dog bones pictured in Figure 2 were manufactured from the aluminium strips and tested in extensometer.

Figure 2: Image of the dog-bones tested in the extensometer.

Three dog bone samples were made, with one failing due to bearing stress so its data was not used in the analysis. Stress-strain curves created from the tensile testing data is displayed in Figure 3.

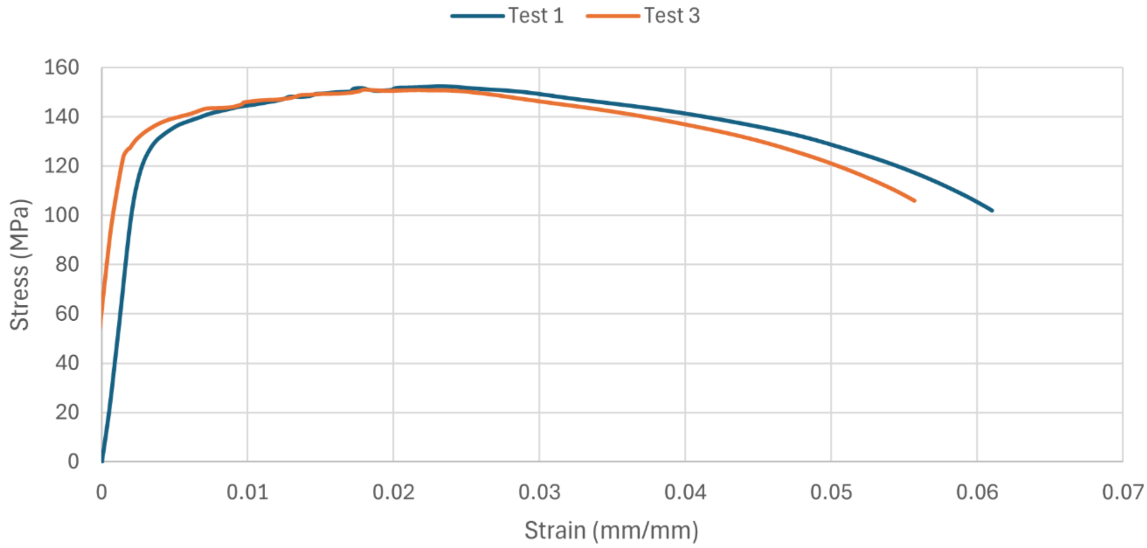


Figure 3: Tensile Testing Stress-Strain Data.

From the data, the ultimate tensile strength and Young's Modulus was obtained for each sample. These values are recorded in Table 1, along with the average values that were subsequently used in further design calculations.

Table 1: Material Properties of Unknown Aluminium.

Sample	UTS (MPa)	Young's Modulus (GPa)
1	151.1	50.7
3	152.4	49.6
Average	151.8	50.2

Design Development

Chosen for their strength and relative simplicity, three-membered triangular structures were chosen to be looked at for initial designs. Three variations pictured in Figure 3 were initially explored.

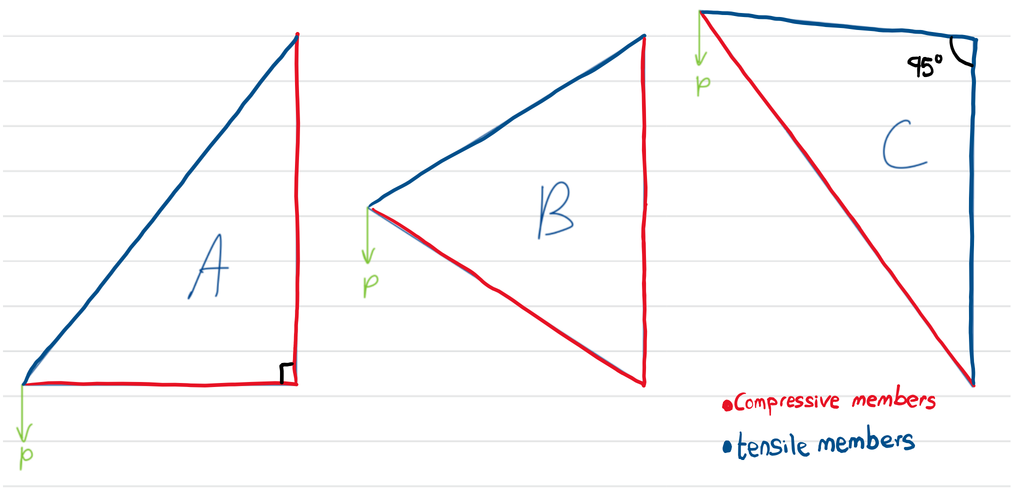


Figure 3: Initial Shape Ideas.

Structures A and B have two compressive members whereas shape C only has one. Compressive members need to be considerably stronger than tensile members to resist buckling, generally making them heavier and more difficult to manufacture. As strength-to-weight ratio is an important consideration in this design, shape C was chosen due to only one compressive member needing to be manufactured. Design C is close to an upper right-angle triangle but has an angle greater than 90 degrees in the top right corner to ensure that the vertical member is in tension.

Failure

Failure Loads

The design requirements specify that the structure must failure due to a load of 20 to 39 kg. To fall well within these requirements, this design aims to fail at 30 kg with a range of $\pm$ 7.5 kg. Free body diagrams were completed along with force calculations to determine the loads through each member while the structure is supporting 30 kg. The FBDs and calculations can be found in the Hand Calculations section of the appendix. Table 2 details the loads acting through each member in the structure.

Table 2: Member Loading while Supporting 30 kg Mass.

Member	Load (N)	Load Type
Main Diagonal	518	Compressive
Vertical	35	Tensile
Horizontal	401	Tensile

Failure Modes

The first failure mode of this design is at a stress concentration in the horizontal tensile member. This was chosen as the preferred mode of failure due to this member having a high tensile load and tensile failure being more predictable than compressive failure.

Two alternative modes of failure are buckling in the compressive member and tensile failure in the vertical member. To design against compressive failure, an I-beam was chosen to be manufactured for the compressive member to resist buckling. The tensile load on the vertical member is relatively small, so no preventative measures were decided to be taken.

A summary of the failure modes is presented in Table 3.

Table 3: Selected Failure Modes

Failure Mode Order	Failure Mode	Associated Member
1st (Preferred)	Tensile Failure at SC	Horizontal
2nd	Buckling	Diagonal
3rd	Tensile Failure	Vertical

Final Design

The final design was modelled in SolidWorks and drawings are presented in the Drawings section of this report. The two tensile members of the structure are single rectangular strips of aluminium, and the compressive member is an I-beam. To reduce the weight of the structure, holes are to be drilled through multiple points in the both the tensile and compressive members. The stress concentration is a notch, chosen for ease of manufacturing, located at the centre of the horizontal tensile member. The required width of the stress concentration to fail at 401 N was calculated to be 2.52 mm.

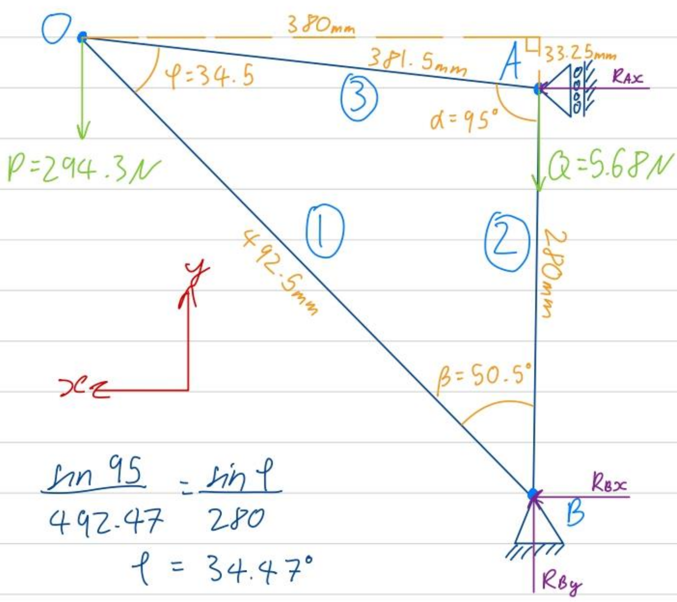
Conclusion

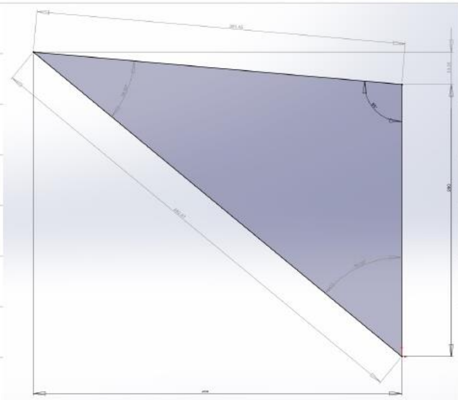
An aluminium structure was designed to fail while supporting a mass $30\ kg$. To determine the material properties of the unknown grade of aluminium used to build the structure, tensile testing was conducted. Force calculations were done for the chosen three-member triangular design, which then informed which modes of failure should be chosen, as well as dimensions for the preferred tensile stress concentration.

Appendix

Hand Calculations

FBD





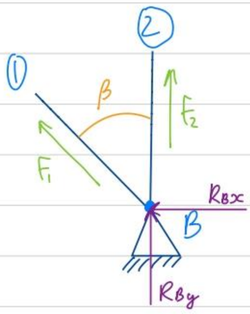
$$\frac{\sin 95}{492.47} = \frac{\sin \varphi}{280}$$
$$\varphi = 34.47°$$

$$\frac{\sin 95}{492.47} = \frac{\sin \beta}{381.45}$$
$$\beta = 50.50°$$

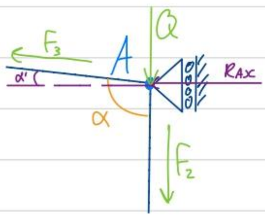
$$\sum M_B = 0 = 0.28 R_{Ax} + 0.38 P$$
$$R_{Ax} = \frac{-0.38P}{0.28}$$
$$= \frac{-(0.38)(294.3)}{0.28}$$
$$R_{Ax} = -399.4N$$

$$\sum F_x = 0 = R_{Bx} + R_{Ax}$$
$$R_{Bx} = -R_{Ax}$$
$$R_{Bx} = 399.4N$$

$$\sum F_y = 0 = R_{By} - P - Q$$
$$R_{By} = P + Q$$
$$= 294.3 + 5.68N$$
$$R_{By} = 300.0N$$



$$\sum F_x = 0 = F_1 \sin \beta + R_{Bx}$$
$$F_1 = \frac{-R_{Bx}}{\sin\beta}$$
$$= \frac{-399.4}{\sin 50.5}$$
$$\boxed{F_1 = -517.6N}$$



$$\sum F_x = 0 = R_{Ax} + F_3 \cos\alpha'$$

$$F_3 = \frac{-R_{Ax}}{\cos\alpha'}$$

$$= \frac{-(-399.4)}{\cos 5}$$

$$\boxed{F_3 = 400.9N}$$

$$\sum F_y = 0 = -F_2 - Q + F_3 \sin\alpha'$$

$$F_2 = -(Q - F_3 \sin\alpha')$$

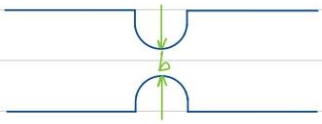
$$= -(5.68 - 400.9 \sin 5)$$

$$\boxed{F_2 = 29.3\ N}$$

The forces in members 1, 2 and 3 are 517.6N(C), 29.3N(T) and 400.9N(T).

Stress Concentration (SC)

SC calculations could be done; however, these would only be relevant for brittle materials. Aluminium is ductile ∴ a SC factor of K=1.1 was determined via physical testing.



$$A = bd$$

$$\sigma_{nom} = \frac{F}{A}$$

$$\sigma_{max} = K\sigma_{nom}$$

$$= K \frac{F}{bd}$$

$$b = K \frac{F}{d\ \sigma_{max}}$$

$d = 1.2mm$

$$= 1.1 \frac{400.9}{(1.2_{E^{-3}})(152_{E^{6}})}$$

$$b = 2.42_{E^{-3}}\ m$$

$$\sigma_{max} = K\sigma_{nom}$$

$$= K \frac{F}{bd}$$

$$b = K \frac{F}{1.15_{E^{-3}}\ \sigma_{max}}$$

$d = 1.15mm$

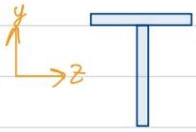
$$= 1.1 \frac{400.9}{(1.15_{E^{-3}})(152_{E^{6}})}$$

$$\boxed{b = 2.52_{E^{-3}}\ m}$$

A notch breadth of 2.52mm is required; however, the depth of the material varies, so during manufacturing the exact value will be measured at the location of the SC and the width will be adjusted accordingly.

Buckling

$E = 50E^{9}\,Pa \quad l = 0.4925m \quad \sigma = 517.6N$



$A = (0.0012)(0.018) = 2.16E^{-5}\,m^2$

$$\bar{y} = \frac{\Sigma(y_i A_i)}{\Sigma(A_i)} = \frac{(9E^{-3})(2.16E^{-5}) + (18.6E^{-3})(2.16E^{-5})}{2(2.16E^{-5})} = 0.0138m$$

$\bar{z} = 0.009m$

$$I_z = \Sigma_i(I_{zi} + d_i^2 A_i)$$
$$= \left(\frac{(18E^{-3})(1.2E^{-3})^3}{(12)} + (4.8E^{-3})^2(2.16E^{-5})\right) + 2\left(\frac{(1.2E^{-3})(18E^{-3})^3}{12} + (4.8E^{-3})^2(2.16E^{-5})\right)$$

$I_z = 1.58112\,E^{-9}\,m^4 = 1581.12\ mm^4$

$$I_y = \Sigma_i(I_{yi} + d_i^2 A_i)$$
$$= \left(\frac{(18E^{-3})(1.2E^{-3})^3}{(12)} + (0)^2(2.16E^{-5})\right) + \left(\frac{(1.2E^{-3})(18E^{-3})^3}{12} + (0)^2(2.16E^{-5})\right)$$

$I_y = 5.85792\,E^{-10}\,m^4 = 585.792\ mm^4$

$$P_{crz} = \frac{C\pi^2 EI}{l^2} = \frac{(1)(\pi^2)(50E^9)(1581.12E^{-12})}{(0.4925^2)}$$

$P_{crz} = 3216.8N$

$$P_{cry} = \frac{C\pi^2 EI}{l^2} = \frac{(\frac{1}{4})(\pi^2)(50E^9)(585.792E^{-12})}{(0.4925^2)}$$

$\boxed{P_{cry} = 297.95N}$

$297.95N < 517.6$

$$FOS = \frac{297.95}{517.6}$$

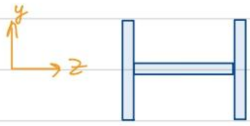
$FOS = 0.576$

∴ T-beam is not suitable for the compressive member.

Buckling

$E = 50E^{9}\,Pa \quad l = 0.4925\,m \quad \sigma = 517.6\,N$

$A = (0.0012)(0.018) = 2.16E^{-5}\,m^2$



$\bar{z} = 0.0102\,m$

$\bar{y} = 0.009\,m$

$$I_y = \sum_i \left(I_{zi} + d_i^{\,2} A_i\right)$$
$$= 2\left(\frac{(18E^{-3})(1.2E^{-3})^3}{(12)} + (9.6E^{-3})^2(2.16E^{-5})\right) + \left(\frac{(1.2E^{-3})(18E^{-3})^3}{12} + (0)^2(2.16E^{-5})\right)$$
$$I_y = 4.56969E^{-9}\,m^4 = 4569.69\,mm^4$$

$$I_z = \sum_i \left(I_{yi} + d_i^{\,2} A_i\right)$$
$$= \left(\frac{(18E^{-3})(1.2E^{-3})^3}{(12)} + (0)^2(2.16E^{-5})\right) + 2\left(\frac{(1.2E^{-3})(18E^{-3})^3}{12} + (0)^2(2.16E^{-5})\right)$$
$$I_z = 1.16899\,E^{-10}\,m^4 = 1168.99\,mm^4$$

$$P_{cry} = \frac{C\pi^2 EI}{l^2}$$
$$= \frac{\left(\frac{1}{4}\right)(\pi^2)(50E^{9})(4569.69E^{-12})}{(0.4925^2)}$$
$$\boxed{P_{cry} = 2324.3\,N}$$

$$P_{crz} = \frac{C\pi^2 EI}{l^2}$$
$$= \frac{(1)(\pi^2)(50E^{9})(1168.99\,E^{-12})}{(0.4925^2)}$$
$$P_{crz} = 2378.3\,N$$

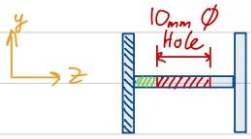
$2324.3 > 517.6$

$$FOS = \frac{2324.3}{517.6}$$
$$FOS = 4.49$$

$\therefore$ I-beam is suitable for the compressive member.

Buckling

$E = 50E^{9}\,Pa \quad l = 0.4925\,m \quad \sigma = 517.6\,N$



$A_1 = (0.0012)(0.018) = 2.16E^{-5}\,m^2$

$A_2 = (0.0012)(0.004) = 4.8E^{-6}\,m^2$

$A_3 = (0.0012)(0.01) = 1.2E^{-5}\,m^2$

$\bar{z} = 0.0102\,m$

$\bar{y} = 0.009\,m$

$$I_y = \sum_i \left(I_{zi} + d_i^2 A_i\right)$$

$$= 2\left(\frac{(18E^{-3})(1.2E^{-3})^3}{(12)} + (9.6E^{-3})^2(2.16E^{-5})\right) + \left(\frac{(1.2E^{-3})(18E^{-3})^3}{12}\right) - \left(\frac{(1.2E^{-3})(10E^{-3})^3}{12}\right)$$

$I_y = 4.46969E^{-9}\,m^4 = 4469.69\,mm^4$

$$I_z = \sum_i \left(I_{yi} + d_i^2 A_i\right)$$

$$= \left(\frac{(18E^{-3})(1.2E^{-3})^3}{(12)}\right) + 2\left(\frac{(1.2E^{-3})(18E^{-3})^3}{12}\right) - \left(\frac{(10E^{-3})(1.2E^{-3})^3}{(12)}\right)$$

$I_z = 1.16755\,E^{-10}\,m^4 = 1167.55\,mm^4$

$$P_{cry} = \frac{C\pi^2 EI}{l^2} = \frac{\left(\frac{1}{4}\right)(\pi^2)(50E^9)(4469.69E^{-12})}{(0.4925^2)}$$

$\boxed{P_{cry} = 2273.4\,N}$

$$P_{crz} = \frac{C\pi^2 EI}{l^2} = \frac{(1)(\pi^2)(50E^9)(1167.55E^{-12})}{(0.4925^2)}$$

$P_{crz} = 2375.4\,N$

$2324.3 > 517.6$

$FOS = \frac{2324.3}{517.6}$

$FOS = 4.49$

∴ the I-beam is still strong enough w/ 10mm Ø holes in the web & the reduction in strength was small so further weight reduction is possible

Strength to Weight (SW)

$V = 4.292629E^{-5}\,m^3$ $\quad \rho = 2710\ kg/m^3$

$W = V\rho$

$= (4.292629E^{-5})(2710)$

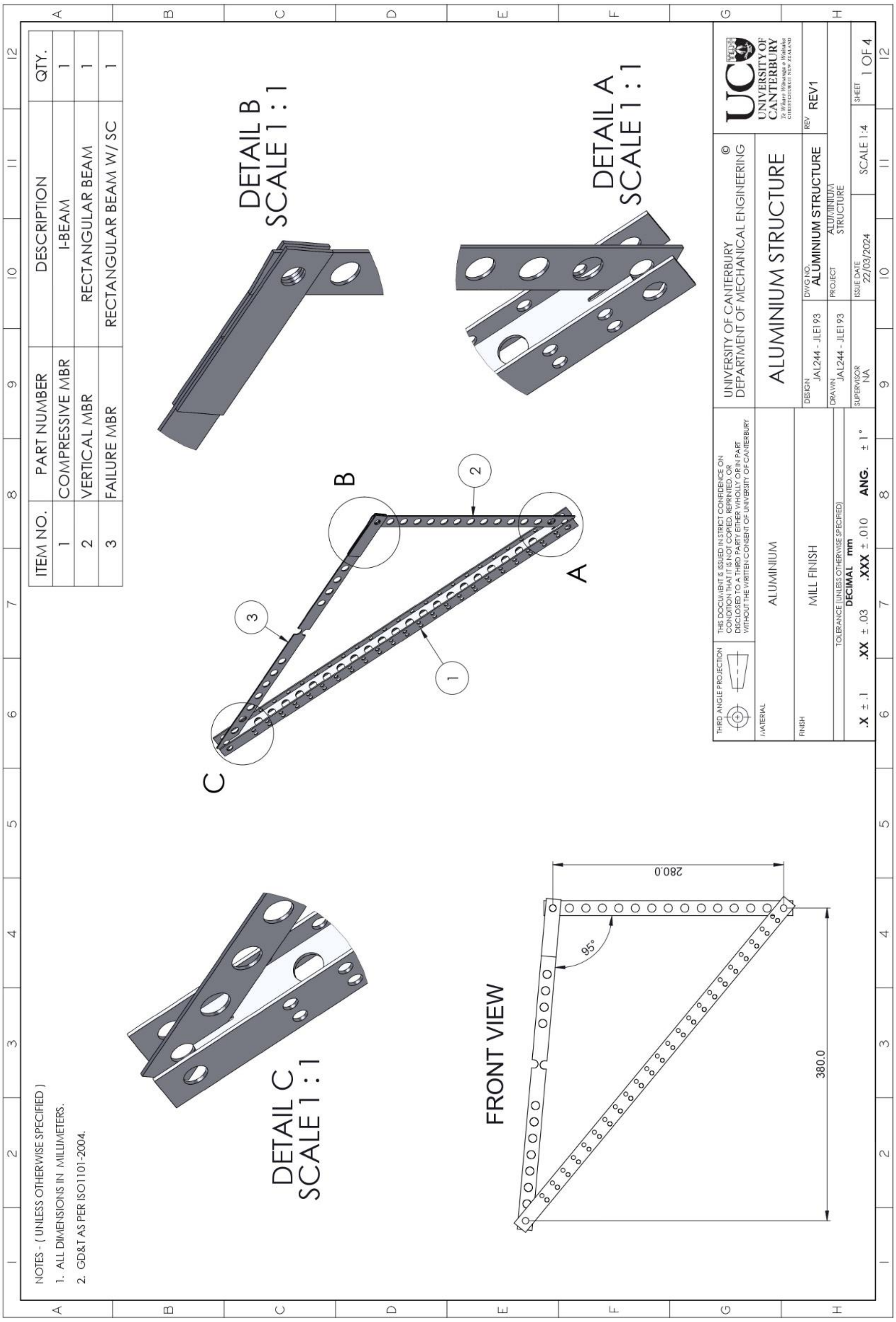
$W = 0.1163\ kg$

$SW = \frac{30}{0.1163}$

$SW = 257.9$

The SW ratio is 257.9; however this does not include the weight of the glue.

Drawings



ITEM NO. | PART NUMBER | DESCRIPTION | QTY.
1 | COMPRESSIVE MBR | I-BEAM | 1
2 | VERTICAL MBR | RECTANGULAR BEAM | 1
3 | FAILURE MBR | RECTANGULAR BEAM W/ SC | 1
DETAIL B SCALE 1 : 1
DETAIL A SCALE 1 : 1
DETAIL C SCALE 1 : 1
FRONT VIEW
A
B
C
1
2
3
280.0
380.0
95°
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FINISH MILL FINISH
TOLERANCE (UNLESS OTHERWISE SPECIFIED) DECIMAL mm
.X ± .1 .XX ± .03 .XXX ± .010 ANG. ± 1°
DESIGN JAL244 - JLE193
DRAWN JAL244 - JLE193
SUPERVISOR NA
DWG NO. ALUMINIUM STRUCTURE
PROJECT ALUMINIUM STRUCTURE
ISSUE DATE 22/03/2024
REV REV1
SCALE 1:4
SHEET 1 OF 4

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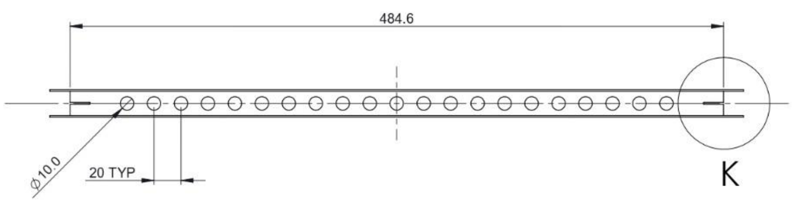
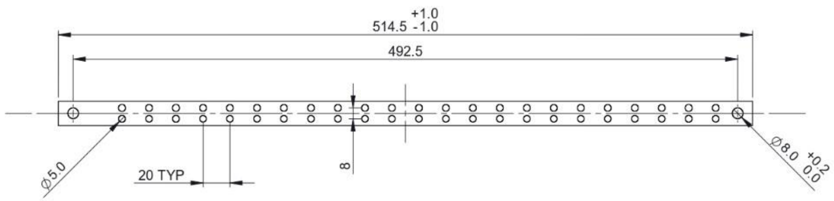
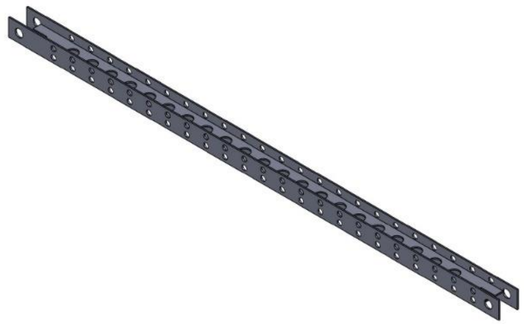
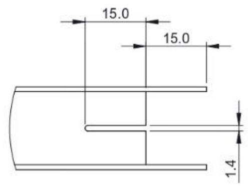
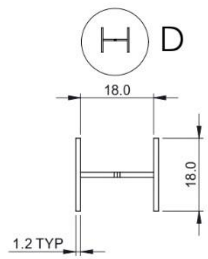








DETAIL E
SCALE 1 : 1



DETAIL D
SCALE 1 : 1

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.X ± .1 .XX ± .05 .XXX ± .010 ANG. ± 1°		SUPERVISOR: G. STILWELL	ISSUE DATE: 23/03/2024 SCALE 1:3	SHEET: 2 OF 4

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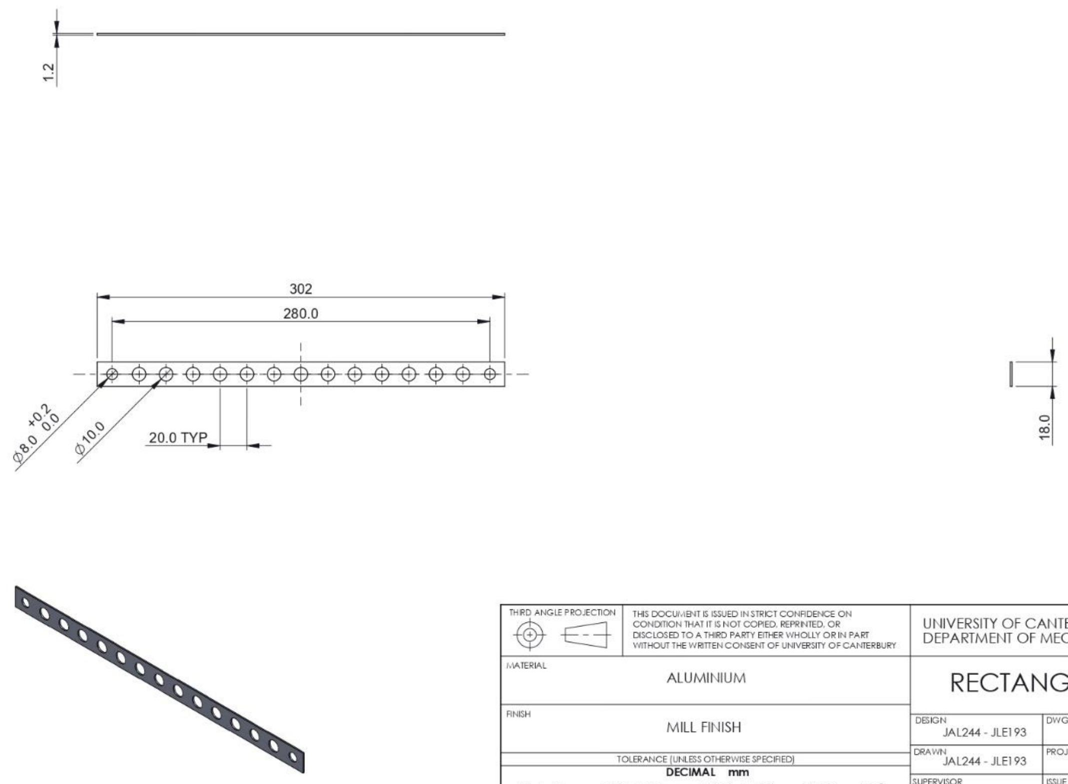

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TOLERANCE (UNLESS OTHERWISE SPECIFIED) DECIMAL mm		DRAWN: JAL244 - JLE193	PROJECT: ALUMINIUM STRUCTURE
.X ± .1 .XX ± .05 .XXX ± .010 ANG. ± 1°		SUPERVISOR: G. STILWELL	ISSUE DATE: 23/03/2024; SCALE 1:3; SHEET: 3 OF 4

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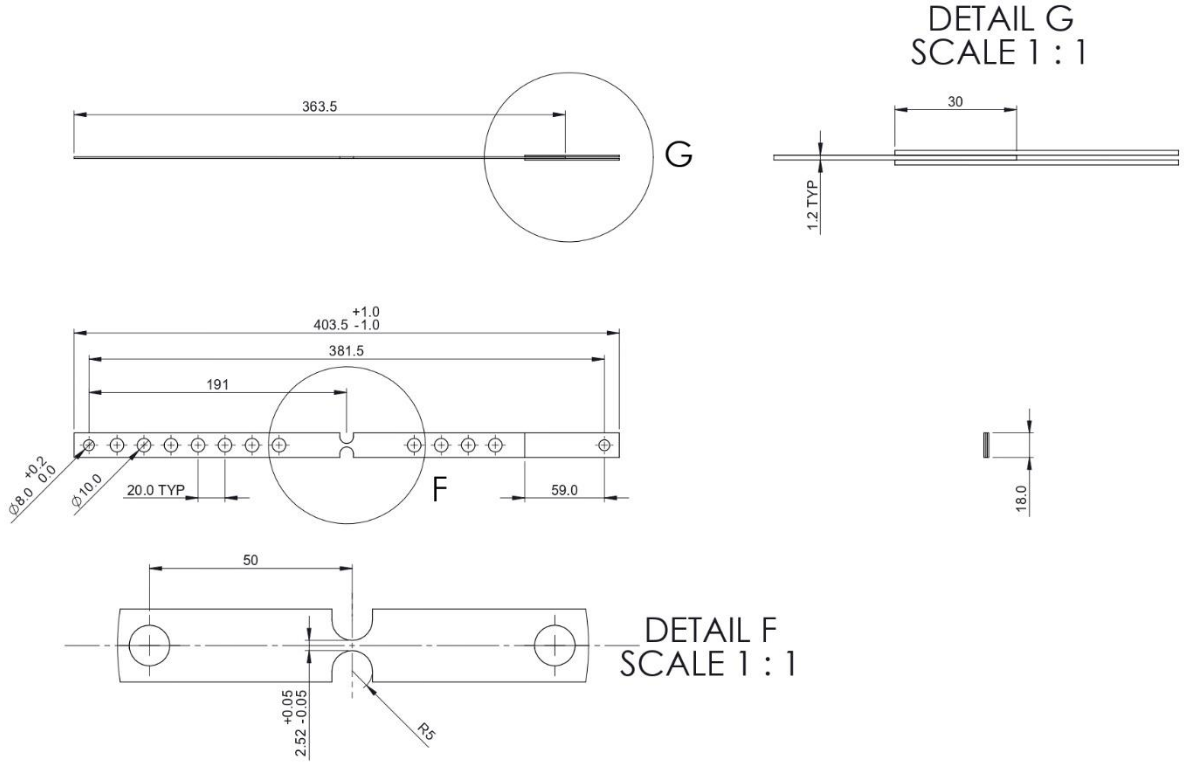

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MATERIAL ALUMINIUM		RECTANGULAR BEAM W/ SC			
FINISH MILL FINISH		DESIGN JAL244 - JLE193	DWG NO. FAILURE MBR		REV REV1
TOLERANCE (UNLESS OTHERWISE SPECIFIED) DECIMAL mm		DRAWN JAL244 - JLE193	PROJECT ALUMINIUM STRUCTURE		
.X ± .1 .XX ± .05 .XXX ± .010 ANG. ± 1°		SUPERVISOR G. STILWELL	ISSUE DATE 23/03/2024	SCALE 1:3	SHEET 4 OF 4

Timesheet

Time Sheet

28/02	8:45pm – 9:15pm	Dogbone design
	2:00pm – 4:00pm	Dogbone manufacture
29/02	9:00am – 10:00am	Dogbone test
05/03	3:00pm – 3:45pm	Data processing
06/03	3:00pm – 3:30pm	Data processing
	3:30pm – 4:30pm	calculations
14/03	3:00pm – 5:00pm	calculations
16/03	7:00pm – 8:30pm	CAD - 3D
20/03	1:00pm – 3:00pm	SC Building e Testing
21/03	2:00pm – 3:00pm	SC Building e Testing
22/03	7:00pm – 8:00pm	CAD -3D
	8:00pm – 12:00pm	CAD - Drawings
23/03	10:00am – 1:30pm	CAD - Drawings
	1:30pm – 2:30pm	Report
	2:30pm – 3:30pm	CAD- 3D / Drawings
24/03	10:00am – 4:00pm	CAD 3D / Drawings / Calcs / report
25/03	11:00am – 3:00pm	Report

Total : 31.75 hours